DSAA 4011 · LAB 7 · FRIDAY 23 OCTOBER 2026

Sharing Data, Not People

Lecture 7's Chow–Liu tree inside private synthetic data

A clinic wants others to train models on its records. Today: how a tree of links lets it share **fake patients** with the same patterns, and no real ones.

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Builds on Lecture 7: Who is linked to whom?

Outline

- **The problem: share the patterns, not the patients**

The protection: noise on every released count

The model: a tree of 2×2 tables

The method: noisy choice, noisy tables, fake patients

What it keeps, what it costs, where it fails

Each part answers the question the last one leaves open.

Main Question

How can a clinic share the patterns in its records without sharing any patient?

Four words for today

Read the clinic's records like this.

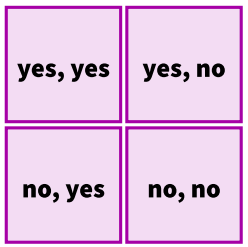
smokes?	coughs?	breathless?	chest pain?
yes	yes	yes	no
no	no	no	no
yes	yes	no	yes
yes	no	no	no
no	yes	yes	no

a question: one column

a patient: one row of answers

a 2×2 table

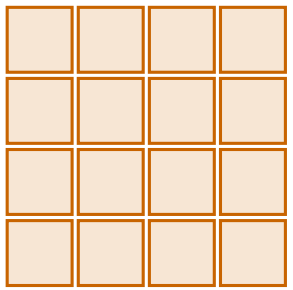
smokes? × coughs?



4 boxes: patients counted by their two answers

the full table

all four questions



16 boxes: one for every combination of answers

SO **patient = a row · question = a column · 2×2 table = counts for two questions.**

NEXT Can the clinic release these rows if it removes the names?

1 · The problem: removing names is not enough

Remove the names, and a public list still finds Bo.

released answers: names removed

born	sex	postcode	smokes	coughs
1999-03-14	F	511453	yes	yes
2001-07-02	M	511400	no	no
1998-11-30	F	511462	yes	no

a public list, with names

name	born	sex	postcode
Ann Li	2001-07-02	M	511400
Bo Wang	1999-03-14	F	511453
Chen Ma	1985-01-09	M	511470

same birth date, sex and postcode

so the first released patient is Bo Wang: a smoker who coughs

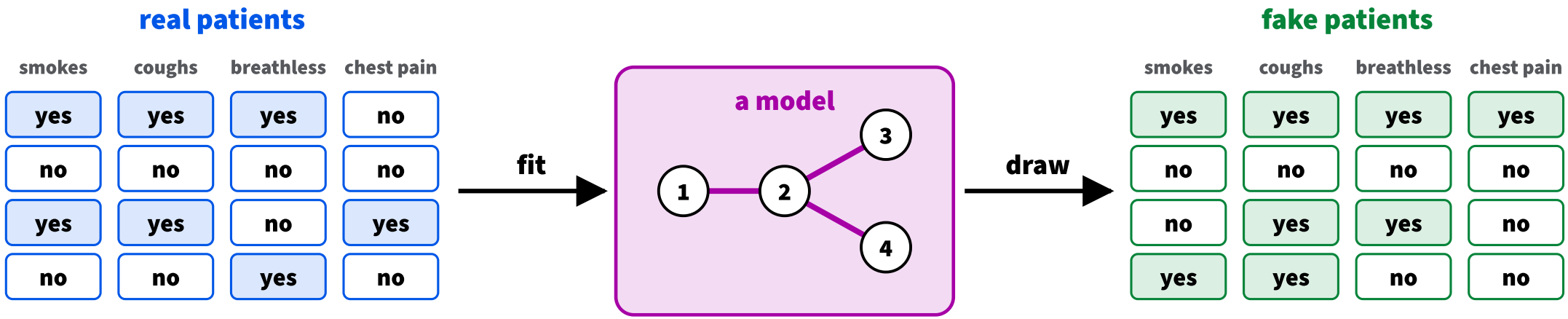
(made-up example)

SO **Real rows cannot be released.**

NEXT Then release fake patients. What must they keep?

1 · The idea: fake patients drawn from a model

Fit a model to the real patients, then draw fake ones from it.



the model stores shared patterns, so no fake patient copies a real one

Q: Draw each question's answer on its own. What is lost? A: The links: smoking no longer predicts coughing.

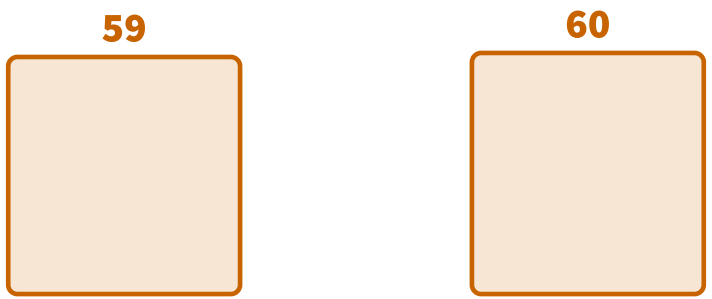
SO **The model must keep the links between questions.**

NEXT A model is built from counts. Is releasing a count safe?

2 · The protection: a count can name one patient

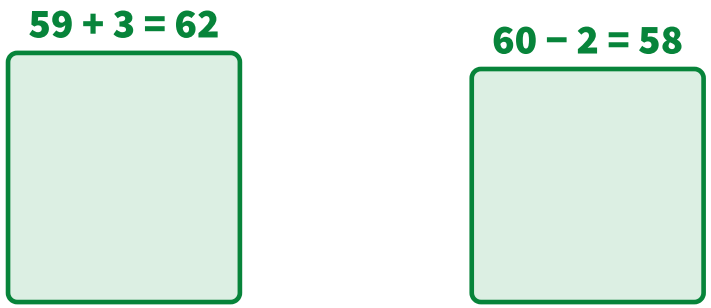
When Bo joins, one count goes up by 1, and that names him.

released as counted



differ by 1: Bo smokes, no cough

released with noise



differ by −4: nothing about Bo

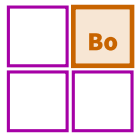
SO **So every released count gets noise: the change of 1 disappears, 59 stays about 59.**

NEXT How much noise does each count need?

2 · The protection: every table is one more clue about Bo

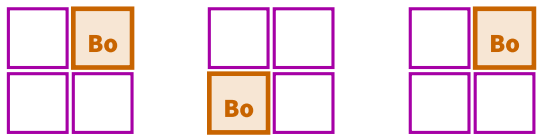
Bo falls into one box of every table we release.

release 1 table



Bo moves 1 count by 1
noise ± 2 hides him

release 3 tables



Bo moves 3 counts, one per table
noise ± 6 on every count

privacy level ε = 1 · the orange box: the one Bo's answers fall into

Clues add up. Release Bo's count 3 times, each with noise ± 2: 62, 57, 61. The average is 60: the noise cancels and Bo's +1 shows again.

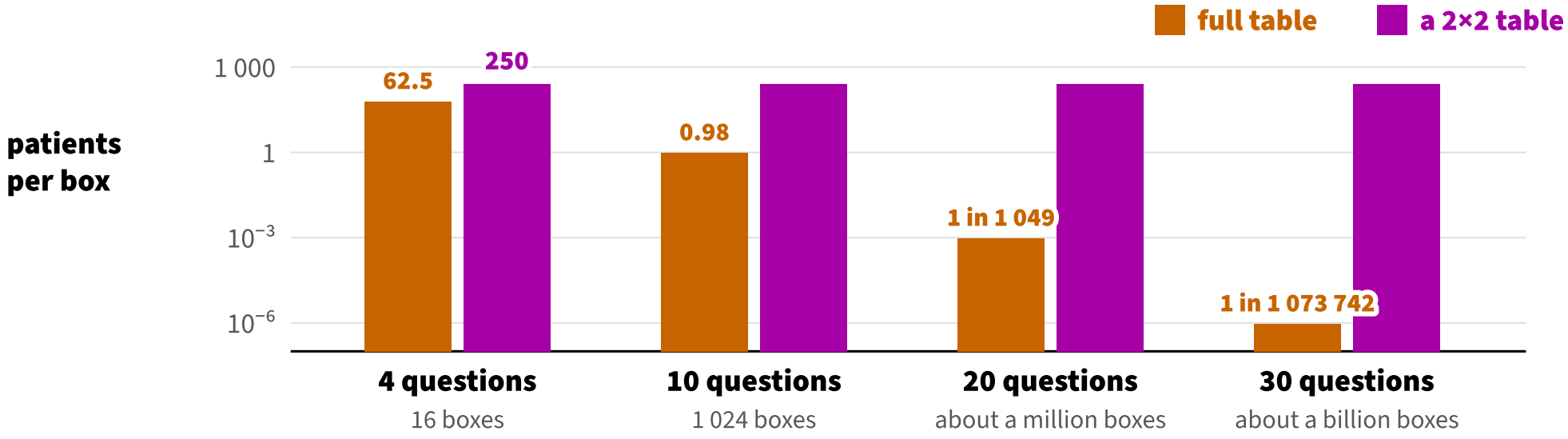
The fix. The privacy level is a fixed budget for all clues together, so 3 tables need ± 6 each, k tables k times the noise.

SO **More tables, noisier counts, worse fake patients: release few tables.**

NEXT Can one table keep all the links?

3 · The model: the full table fails

One table for all questions has 2^m boxes; 1000 patients spread over them.



SO **20 questions: a million boxes, almost all 0 or 1, buried in noise. A 2×2 table's boxes hold 250 each on average.**

NEXT Few 2×2 tables have big counts. Can they keep the links?

3 · The model: a tree passes both tests

Compare the three choices on the two tests from Parts 1 and 2.

what the clinic releases	keeps the links?	big counts, safe under noise?
each question alone (4 counts)	✗ links lost	✓
the full table (2^m boxes)	✓	✗ boxes hold 0 or 1 patient
a tree (m − 1 tables of 2×2)	✓ the strongest links	✓ about 250 patients per box

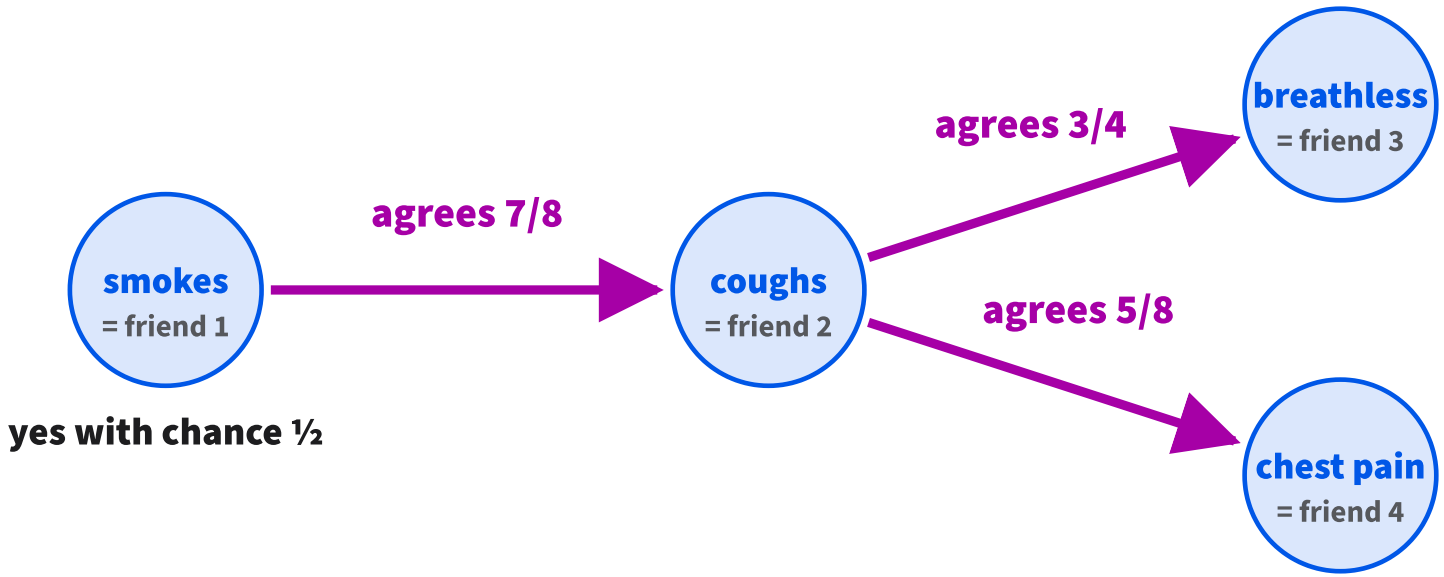
And a tree is easy to draw from: answer one question, then follow the arrows.

SO **A tree: m − 1 tables, big counts, the strongest links.**

NEXT Which tree, for our clinic?

3 · The model: the clinic's tree

Lecture 7's tree: questions are the friends, patients are the days.



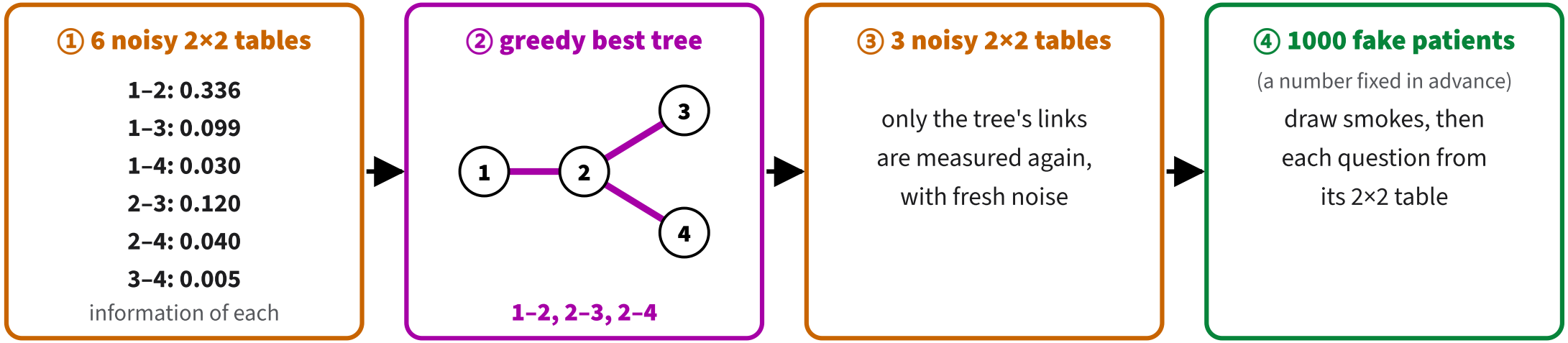
Lecture 7's friends became questions, days became patients: the same tree, the same numbers

SO **A fake patient: answer smokes, then follow the arrows. 7 numbers in all.**

NEXT Choosing the tree also reads the data. Is the choice safe?

4 · The method: protect the choice, then the tables

Step ① gives all 6 pairs noise, so the choice of tree is protected too.



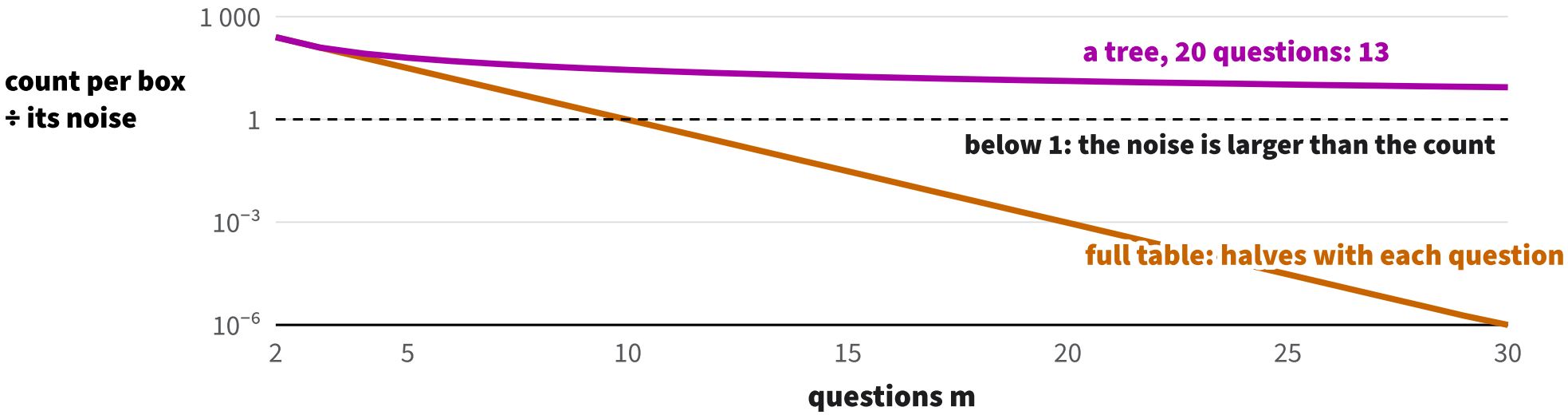
questions: 1 smokes, 2 coughs, 3 breathless, 4 chest pain · privacy: half on ①, half on ③

SO **Every count leaves with noise; the fake patients use only the tree's 3 noisy 2×2 tables.**

NEXT With so much noise, are the counts still useful?

4 · The method: counts stay above the noise

A count is useful only if it is larger than its noise.

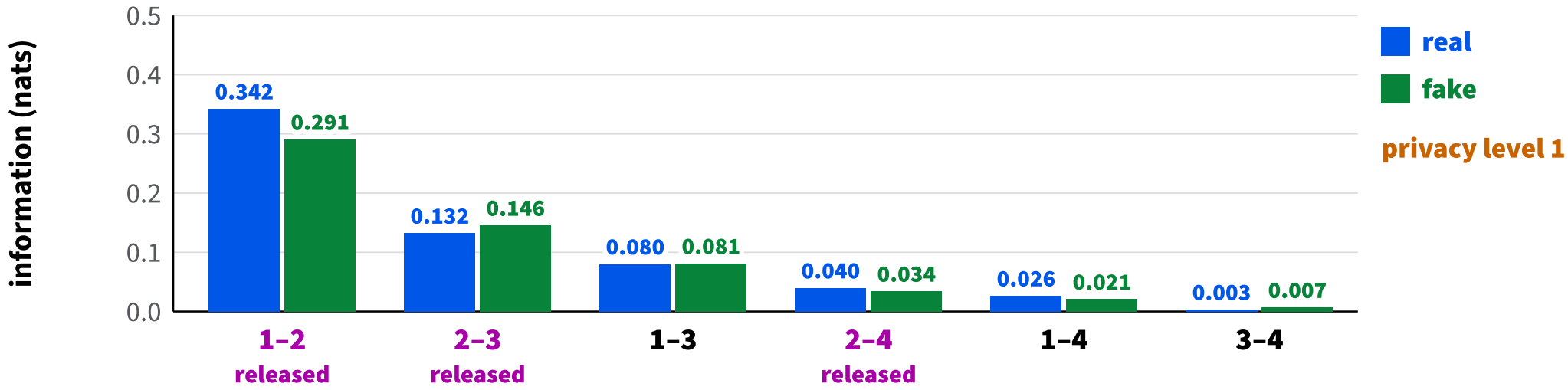


SO **The tree's step ③ counts stay far above the noise; the full table's sink below it from 10 questions.**

NEXT Release the clinic: are the links kept?

5 · The result: every link kept

The fake patients keep every link, even ones never released.



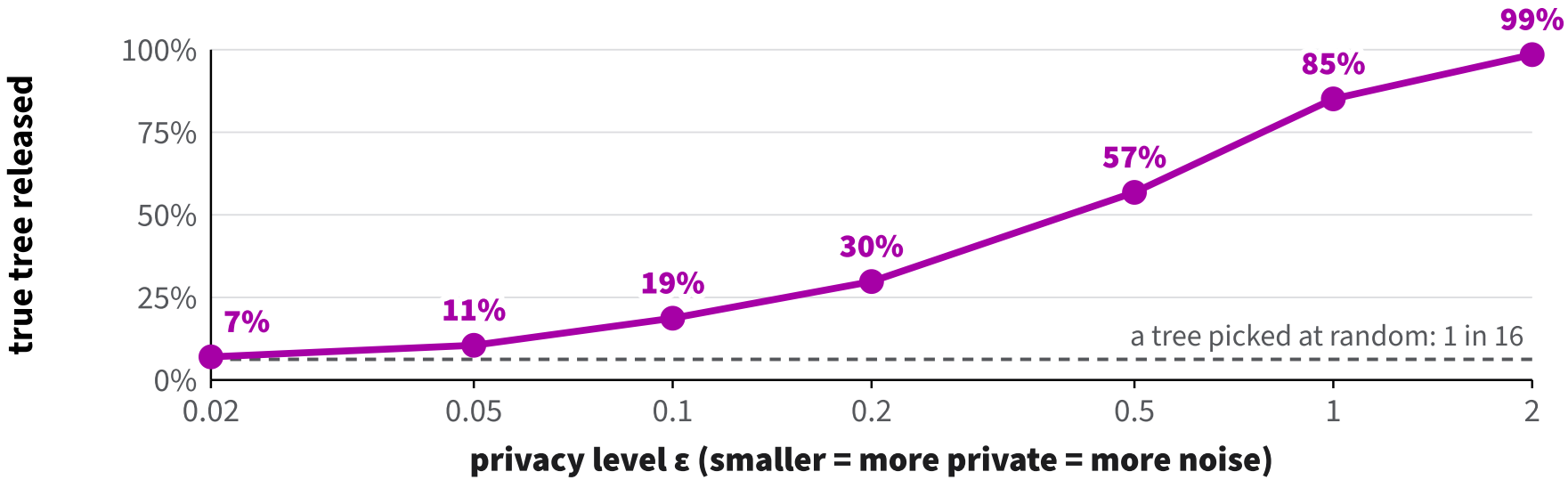
Q: No table for smokes and breathless (pair 1–3) was released, yet the fake data keeps its link: 0.081 against 0.080. Where does it come from? A: From the path smokes → coughs → breathless: a copy of a copy.

SO **Right tree, every link kept.**

NEXT What if we ask for more privacy?

5 · The cost: more privacy, wrong trees

More privacy means more noise, and more wrong trees.

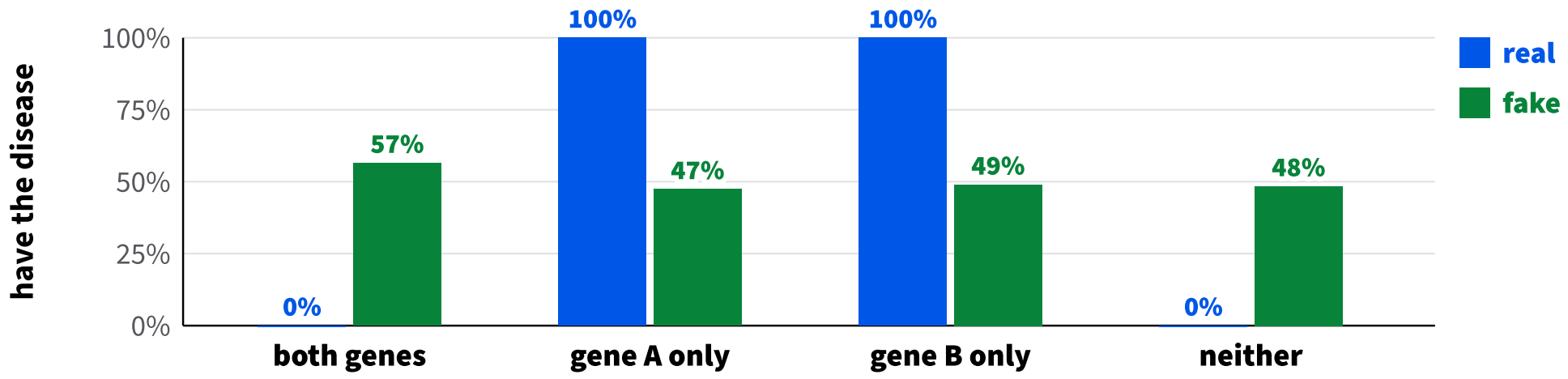


SO **Right in 85% of releases at ε = 1, only 19% at ε = 0.1: weak links go first.**

NEXT Even without noise, can a tree miss a pattern?

5 · The limit: a pattern no pair can see

A disease appears when exactly one of two genes is present.



every pair of questions: information 0.001, 0.000, 0.000 (no pair is linked)

Q: Every pair of questions has information about 0. In the fake patients, what share of "gene A only" patients will have the disease? A: About half, as in every group: the pattern is gone.

SO **A tree keeps only what pairs can see.**

NEXT Where do these ideas appear in today's artificial intelligence (AI)?

5 · In AI today: the clinic's four ideas, at scale

Each idea of this lab reappears in today's artificial intelligence (AI).

idea in the clinic	in AI	example
fit a model, draw fakes	a generative model: a language model draws one word at a time, as the tree draws one answer	every large language model
fake patients for others	private synthetic data to **train models** on	MST (maximum spanning tree method) won the 2018 contest of NIST, the US National Institute of Standards and Technology; AIM (adaptive and iterative mechanism) adds bigger tables
a count can name Bo	a model can **memorize** a training record	names and phone numbers pulled out of GPT-2 (OpenAI's Generative Pre-trained Transformer 2)
noise on every count	noise on every **gradient** while training a network	DP-SGD (differentially private stochastic gradient descent), with the same privacy level ε

SO **Generative AI learns a distribution and samples it; noise keeps the Bo in its data hidden.**

NEXT The argument in three sentences.

Conclusion

TROUBLE Rows without names **still leak people**, and even one count can name a patient.

FIX Add noise to every count, release only a **tree of 2×2 tables**, then draw fake patients along it.

RESULT The links survive, indirect ones too; more privacy costs weak links first; patterns that pairs cannot see are lost.

Remember: fake data keeps only the links the released tables can see.

Your turn

Predict, then release the clinic.

Experiment 1: the clinic (`python lab7_explore.py 1`, course site → Week 7 → Lab files)

- Privacy level 0.3 instead of 1: is the tree still right?
- 10 000 patients instead of 1000: at ε = 0.1, does the tree come back?

Experiments 2 and 3

2. The two-gene disease: also release the three-question table. Does the pattern return?
3. More questions: full table against tree, from 4 to 16 questions.

Tip: write your guess first; surprises teach.

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Thank you!

Next: the midterm on Thursday 29 October, on Lectures 1 to 6

Backup: privacy in one line, why k tables need k times the noise, the strong-privacy release, the real MST.

Backup: differential privacy in one line

The release barely changes when any one patient is removed.

$$\Pr\left(\text{release} \in A \mid D\right) \;\le\; e^{\varepsilon} \cdot \Pr\left(\text{release} \in A \mid D'\right)$$

- for every set A of releases and any two record sets D, D' differing by one patient, in either order;
- **Laplace noise** with scale b has density $\propto e^{-|z|/b}$; if removing one patient changes the counts by at most k in total, scale $b = k/\varepsilon$ gives level ε;
- two releases at levels ε_1 and ε_2 together have level $\varepsilon_1 + \varepsilon_2$: our two halves add to ε, even when the second depends on the first;
- anything computed only from the noisy counts (the tree, negatives set to 0, the fake patients) keeps level ε.

Read: small ε, the two worlds look alike, so no one can tell whether a given patient was in the records.

Backup: why k tables need k times the noise

One patient sits in one box of every released table.

remove one patient: each of the k tables loses 1 in one box

> so the counts change by k in total

noise scale per count $b = \dfrac{k}{\varepsilon}$

> Laplace noise for level ε

choosing the tree: $k = 6,\ \varepsilon/2 \ \Rightarrow\ b = \dfrac{6}{\varepsilon/2} = \dfrac{12}{\varepsilon}$

> all six 2×2 tables, half the privacy

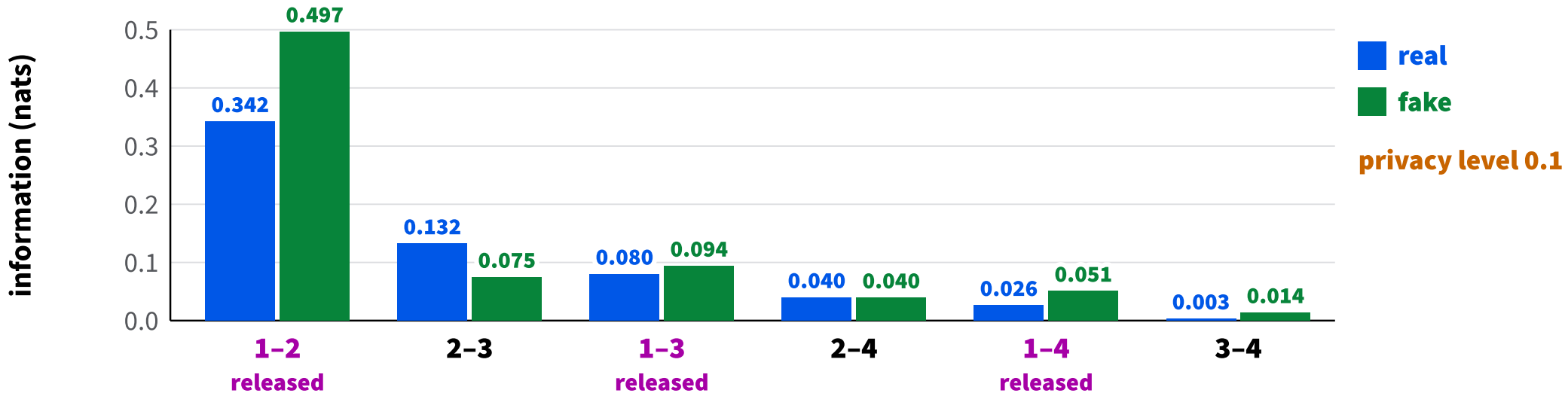
the tree's tables: $k = 3,\ \varepsilon/2 \ \Rightarrow\ b = \dfrac{3}{\varepsilon/2} = \dfrac{6}{\varepsilon}$

> three tables, the other half

Notice: at ε = 1 the tree's counts get noise of about ± 6 against counts from 53 to 456; at ε = 0.1, about ± 60.

Backup: one release at privacy level 0.1

Strong noise links smokes to every question.



Read: the released tree is 1–2, 1–3, 1–4, not 1–2, 2–3, 2–4; in the fake data coughs and breathless keep only 0.075 of their 0.132; the noisy smokes–coughs table lost its "smokes, no cough" box, so 631 fake patients cough against 509 real.

Backup: the real MST, step by step

MST chooses the tree inside the greedy rule, privately.

1. Measure every question's own counts, with noise.
2. Weigh each pair by how far its counts are from "no link" (a stand-in for its information).
3. Run the greedy rule, but at each step **pick a strong link at random**, stronger links more likely: Kruskal's algorithm made private.
4. Measure the tree's 2×2 tables with noise; fit one model to all measurements, then draw fake records.

Read: it spends a third of its privacy on choosing the tree and two thirds on measuring; our lab version replaces steps 1–3 by noisy tables of all pairs and the plain greedy rule, with half of ε.